Bayesian meta-analysis with *baggr*

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baggr

baggr (Bayesian aggregator) is a new R package for Bayesian meta-analysis models that uses Stan

baggr beta is available on CRAN and GitHub

Motivation and goals

Our goals:

1. Implement all basic Bayesian meta-analysis models and tools
2. Focus on accessibility, model criticism and comparison
3. Help people avoid basic mistakes
4. Keep the framework flexible and extend to more models

(Probably) *not* our goal:

5. Build a package for people who already build their models in Stan

Walk through a simple example (8 schools!)

After Rubin (1981)[1]

- Assume only aggregate data available
- Bayesian hierarchical model. Each study measures pair (y_i, se_i):
  - $y_i \sim \mathcal{N}(\tau_i, se_i)$
- Upper level:
  - $\tau_i \sim \mathcal{N}(\tau, \sigma)$
- $\sigma = 0$ is "full pooling", $\sigma = \infty$ is "no pooling"; otherwise partial pooling.
- We need a prior on σ, τ; se_i treated as known.

[1] Rubin, Donald B. "Estimation in Parallel Randomized Experiments". Journal of Educational Statistics 6, no. 4 (1981): 377–401.

Why go Bayesian with meta-analysis?

Beyond "vanilla" arguments. In the package:

- Incorporate prior beliefs (uniquely suitable to meta-analysis)
- Avoid problems with underestimated variance
- Emphasis on uncertainty (also in heterogeneity)
- Interpretability and ease of creating predictive distributions
- Work with derived quantities (e.g. $\Pr(\forall_i : \tau_i > x)$)
- Easy model comparison

Cost of this: computation, model checking, higher technological barrier to entry etc.

- Flexible (code your model "by hand")

Starting analysis

- You need `rstan` and `baggr` packages in R.
- No extra software needed, but you need RTools.

```
library(baggr)
schools
```

```
##      group tau se
## 1 School A  28 15
## 2 School B   8 10
## 3 School C  -3 16
## 4 School D   7 11
## 5 School E  -1  9
## 6 School F   1 11
## 7 School G  18 10
## 8 School H  12 18
```

(Code for individual-level data would look almost identical.)

baggr syntax example: simple model workflow

```
# Fitting function is baggr()
fit <- baggr(schools, model = "rubin", pooling = "partial",
             prior_hypermean = normal(0, 25), prior_hypersd = uniform(0, 100))

# Semi-automatic prior choice, automatic model choice (with prompts):
eff_lab <- "mean SAT improvement"
fit_c <- baggr(schools, prior_hypersd = cauchy(0, 2.5), effect = eff_lab)
fit_n <- baggr(schools, prior_hypersd = normal(0, 5), effect = eff_lab)
fit_u <- baggr(schools, prior_hypersd = uniform(0, 100),
                        prior_hypermean = normal(-5, 10), effect = eff_lab)

# Plot results for one model, making use of bayesplot
baggr_theme_update(text = element_text(family = "mono"))
plot(fit_c, style = "areas", hyper = TRUE, order = TRUE)

# Compare fit to data
bgc <- baggr_compare("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u)
bgc; plot(bgc)
# Compare posterior predictive distribution (note ggplot)
effect_plot("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u) + xlim(-10, 30)

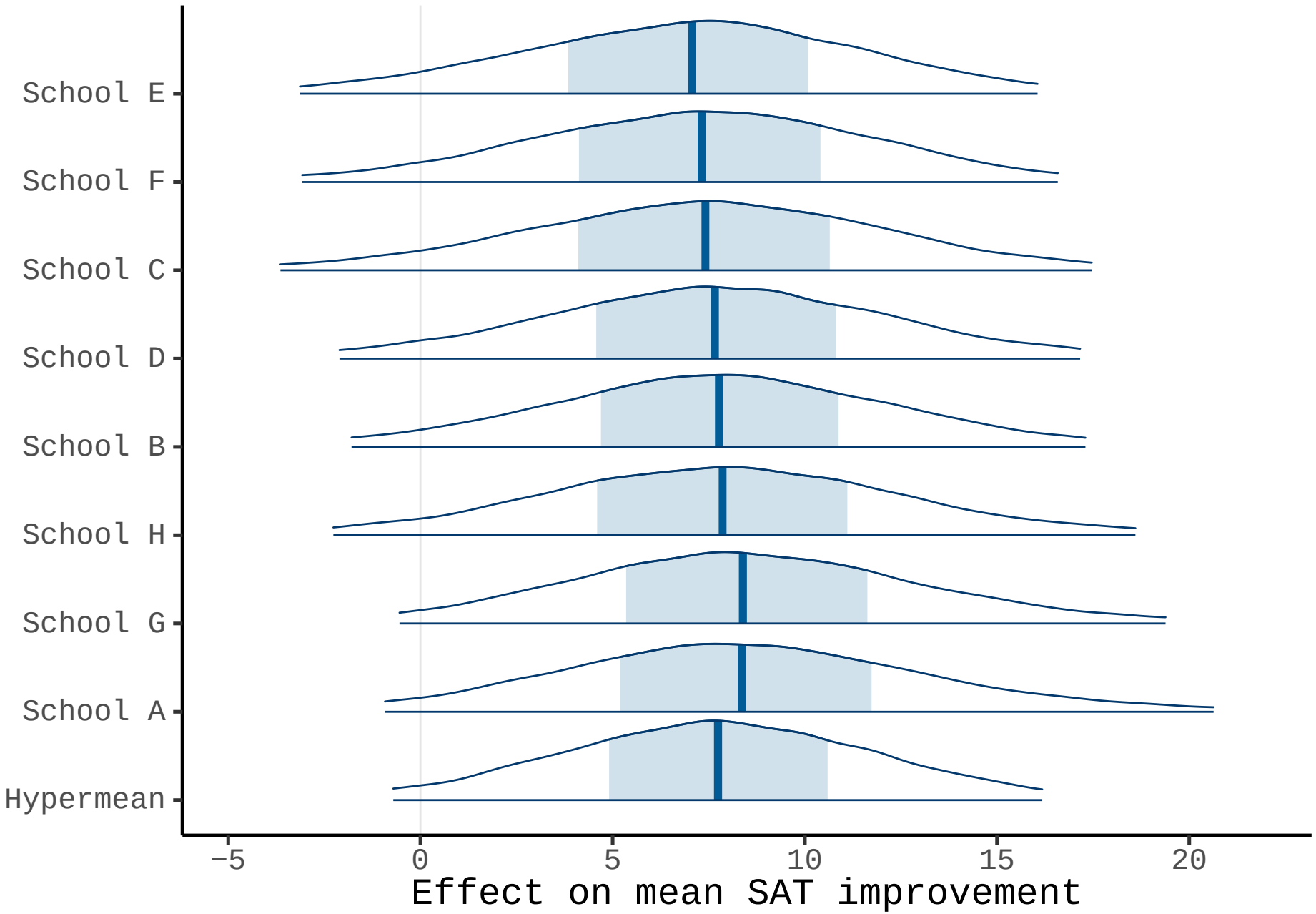
# Compare the impact of different prior choices
bgc_pooling <- baggr_compare(schools, prior_hypersd = normal(0, 10), what = "pooling")
plot(bgc_pooling)

# Use LOO cross-validation to choose between partial & full pooling
loo_p <- loocv(schools, pooling = "partial")
loo_f <- loocv(schools, pooling = "full")
loo_compare(loo_p, loo_f)
```

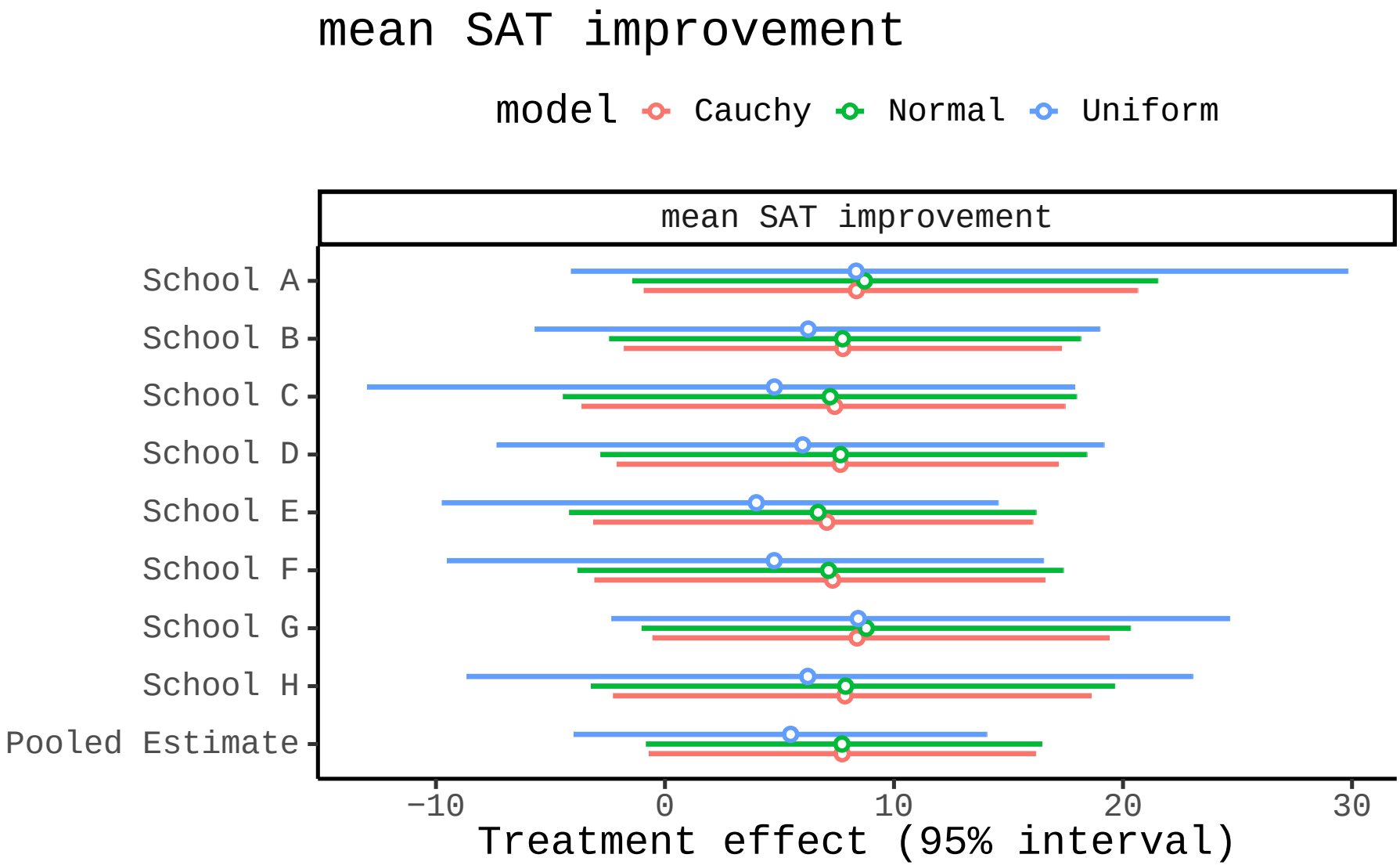
```
fit_c
```

```
## Model type: Rubin model with aggregate data
## Pooling of effects: partial
##
## Aggregate treatment effect (on mean SAT improvement), 8 groups:
## Hypermean (tau) =  7.75 with 95% interval -0.71 to 16.18
## Hyper-SD (sigma_tau) = 2.580 with 95% interval 0.084 to 9.580
## Total pooling (1 - I^2) = 0.93 with 95% interval 0.60 to 1.00
##
## Group-specific treatment effects:
##          mean  sd  2.5% 50% 97.5% pooling
## School A  8.7 5.5 -0.92 8.4    21    0.95
## School B  7.8 4.8 -1.79 7.8    17    0.91
## School C  7.3 5.3 -3.64 7.4    17    0.96
## School D  7.7 4.9 -2.11 7.7    17    0.93
## School E  6.9 4.9 -3.13 7.1    16    0.90
## School F  7.2 4.9 -3.08 7.3    17    0.93
## School G  8.6 5.0 -0.54 8.4    19    0.91
## School H  7.9 5.3 -2.26 7.9    19    0.97
```

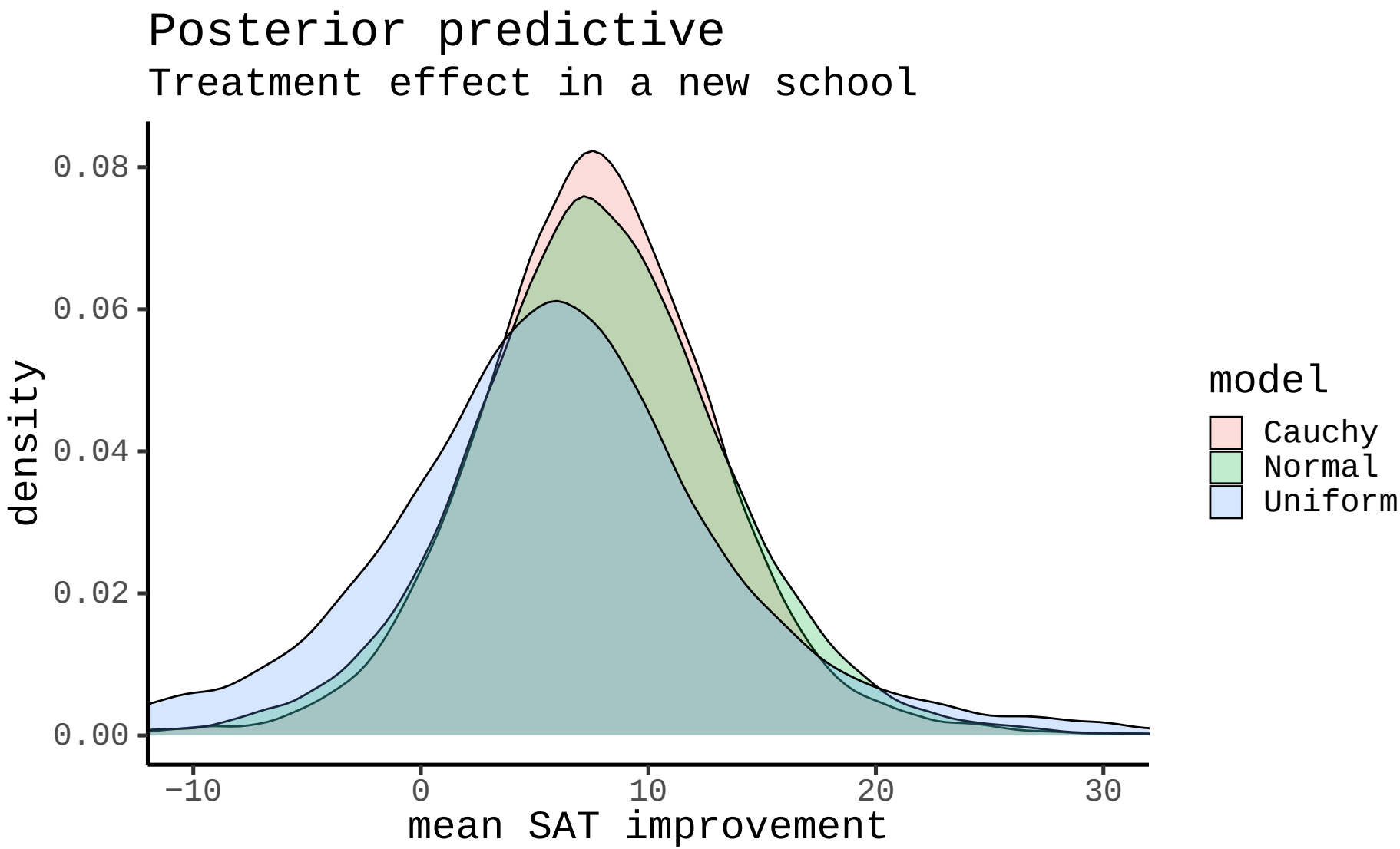
```
plot(fit_c, style = "areas", hyper = TRUE, order = TRUE)
```



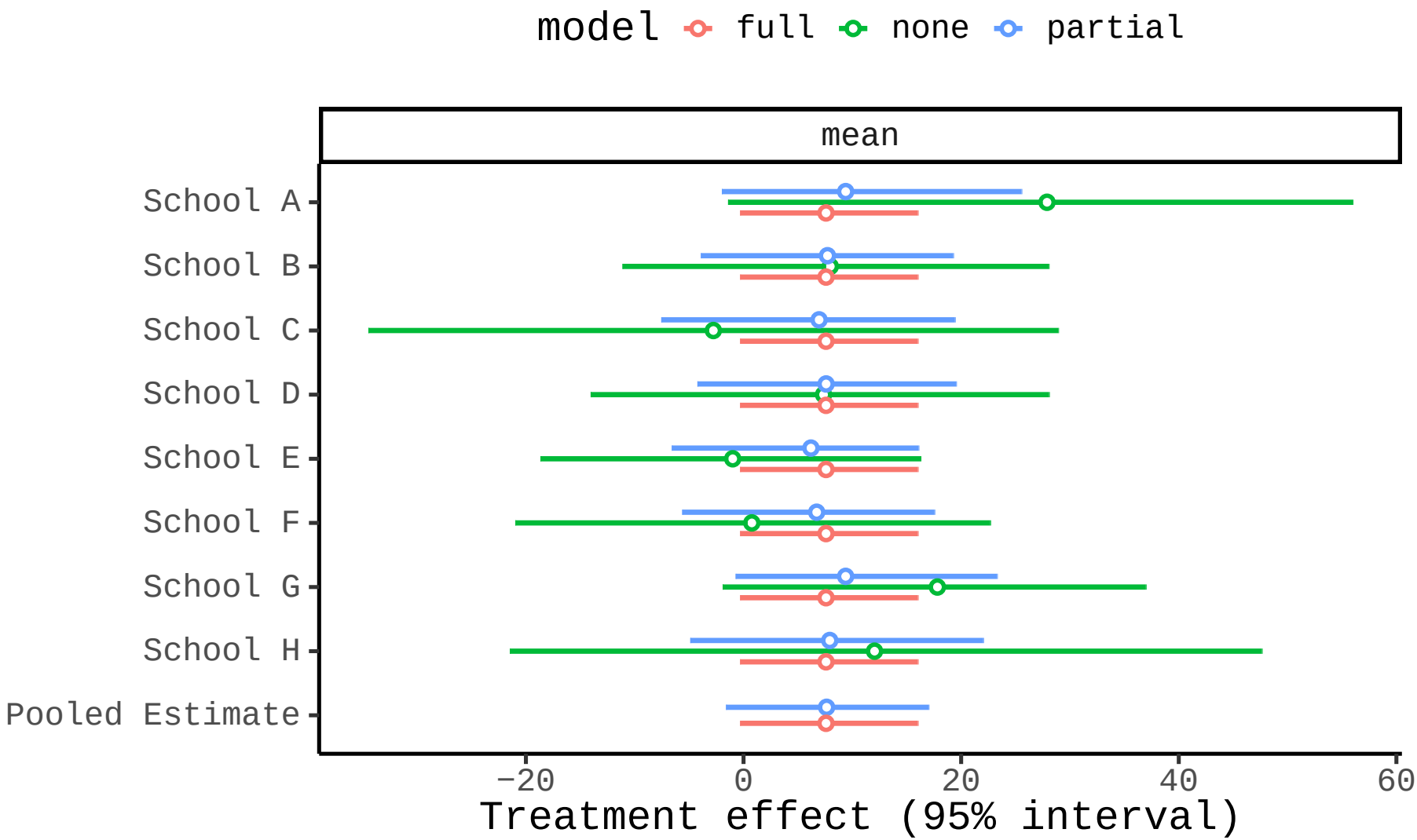
```
bgc <- baggr_compare("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u)
bgc; plot(bgc)
```



```
effect_plot("Cauchy" = fit_c, "Normal" = fit_n, "Uniform" = fit_u) +
  coord_cartesian(xlim = c(-10, 30)) +
  ggtitle("Posterior predictive", "Treatment effect in a new school")
```



```
bgc_pooling <- baggr_compare(schools, prior_hypersd = normal(0, 10),
                             what = "pooling")
plot(bgc_pooling)
```



```r
loo_p <- loocv(schools, pooling = "partial")
loo_f <- loocv(schools, pooling = "full")
loo_compare(loo_p, loo_f)
```

```
## Comparison of cross-validation
##
##                       ELPD ELPD SE
## Model 1 - Model 2 -0.663   0.374
```

Additional commands:

```r
treatment_effect(fit)
group_effects(fit)
effect_draw(fit)
forest_plot(fit)
pooling(fit)
```

Thank you

baggr on GitHub: `github.com/wwiecek/baggr/`